

CONCLUSION

This project aimed to develop an AI model for detecting diseases in banana plant leaves using a self-collected dataset categorized into five classes: Black and Yellow Sigatoka healthy, potassium_deficiency, and panama_disease. Six Deep Learning model (CNN, EfficientNet, MobileNetV2, DenseNet, InceptionV3, and ResNeXt) were trained and evaluated based on accuracy, precision, recall, F1-score, and confusion matrices. Among them, ResNeXt achieved the best performance, followed by EfficientNet.

The project demonstrated the potential of AI in automating disease detection for banana crops, providing a valuable tool for farmers. Further improvements, such as fine-tuning and real-time deployment, are planned for future work.